

CRITICAL REVIEW AND INVITED COMMENTARY

Pertussis vaccination and epilepsy—an erratic history, new research and the mismatch between science and social policy

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SUMMARY

For over 50 years, concerns have been raised about the risk of pertussis vaccine-induced childhood encephalopathy and epilepsy. This article reviews the scientific literature, and the social and historical context in which the scientific, public health and societal views have not always been aligned. Large-scale studies of this issue have produced conflicting results, although the recent consensus is that the risk of vaccine-induced encephalopathy and/or epilepsy, if it exists at all, is extremely low. Risk estimates in the literature have included: risk of a febrile seizure 1 per 19,496 vaccinations; risk of an afebrile seizure 1 per 76,133 vaccinations; risk of encephalopathy after pertussis infection nil—3 cases per million vaccinations. A recent study

showed that encephalopathy in 11 out of the 14 children studied, although previously attributed to vaccination, was in fact due an inherited genetic defect of the *SCN1A* gene that codes for the voltage gated neuronal sodium channel. This study is important because it provides a solid alternative explanation for the perceived pertussis vaccine-encephalopathy association. The interesting possibility is raised that the encephalopathy apparently due to pertussis itself may, in some cases, be due to an *SCN1A* mutation. It may also, by analogy, shed some light on the continuing debate about other serious long-term adverse effects of vaccination in general.

KEY WORDS: Pertussis vaccination, Encephalopathy, Epilepsy, Postvaccination encephalopathy.

Perceptions of risk are often not rational. Many are afraid to travel in airplanes for fear of an accident, a terrorist attack, yet these same people may smoke or not routinely wear safety belts while driving. Dispassionate assessment of competing risks based upon objective scientific principles is muddled especially when governmental and industrial agendas are perceived to threaten the rights, health, or belief-structures of individuals. Occasional abuses of power—whether well intended or not—do little to instill confidence and trust. Perhaps nowhere is the conflict more heated than when it comes to matters of our personal health. Eugenics, fluoridation of public water supplies, HPV vaccination for young girls, bioengineered crops, and the complex debate over stem cell research are a few examples that have lead to a public misunderstanding

and distrust of science and government, especially when the two are mixed. Sound science may be discounted as propaganda, and junk science elevated to the stature of indelible truth. Emotions tend to rule the day.

Mandatory childhood vaccination is another area in which politics and science are inexorably intertwined, and in which the medical progress has fallen foul of societal opinion and become entangled in the legal process. This has always been the case, even though vaccination is probably the medical intervention that has saved more lives and prevented more disability than any other. Its success has been unparalleled—in England for instance, smallpox deaths fell in 1850–1869 from 196 per million to 26 per million in 1885–1894 after only 20 years of a vaccination campaign (Glynn & Glynn, 2005), and this disease which was the scourge of the world has now been completely abolished (at least in the wild, if not in governmental/scientific stockpiles). Similar extraordinary successes are recorded for polio and measles. Public opinion, however, has often been vocal in its opposition to vaccination. Even in 1914 it could be written: “Many of us would prefer an attack of smallpox to vaccination, all of us

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would prefer the risk of smallpox to the certainty of vaccination. Smallpox is a natural disease running a known course Vaccination is a loathsome disease of uncertain origin, artificially transmitted, through various beasts and capable of setting up a variety of repulsive, dangerous and even fatal affections" (Millard, 1914). The reaction to swine flu vaccine is another illuminating example, as recorded in a feature article in the Washington Post in 1976, "An injection is nothing to smile about. . . for too long we have believed uncritically in science, swallowing whole what we are told. Sometimes science wrapped itself in the flag and was called 'American knowledge' . . . so when it comes to swine flu vaccine, I feel cynicism is healthy. Any program conceived by politicians and administered by scientists comes to us doubly plagued" (Cohen, 1976). Herein were contained many of the subsequent lines of argument which continue to influence public opinion against vaccination in general: coercion (especially by politicians/scientists/pharmaceutical industry), un-naturalism, spiritual pollution, ignorance about side effects, commercial greed and the suspicion that vaccination, like eugenics and fluoridation, is a devious political plot. These opinions are still prominent.

Pertussis vaccination is an example case of all of these contrary tendencies; this scientific development has been mired in political, financial, legal, and social controversy for much of its history, and now has a new twist in the tale. It seems timely in *Epilepsia* to review the troubled history of this effective but problematic treatment, and its current position in relation to epilepsy.

In the prevaccination era, pertussis (whooping cough) was a highly prevalent and dire disease, with a significant mortality in young children. In the United States for instance, between 1922 and 1931, about 17 million cases of whooping cough were reported and there were 73,000 deaths. Its infectivity is high with attack rates in nonimmunized households between 70% and 100%. Furthermore, it has been estimated to cause encephalopathy in about 1 out of every 1,200 to 12,000 infections (U.S. Institute of Medicine, 1991; Committee on Infectious Diseases, 1996; Geier & Geier, 2002). The risk of encephalopathy following pertussis is the highest in neonates and infants. Of these facts, there has been little dissension.

Work on vaccine preparation was begun as early as 1906, and indeed by 1914, there were six pertussis vaccine manufacturers. These early vaccines, however, were of limited clinical value and their safety was unclear. Madsen (1933) for instance recorded two deaths "one half hour after the last injection, contractions in the arms and legs occurred followed by cyanosis, hiccup, convulsions and death within a few minutes." Although such cases probably were anaphylactic, the question of a "vaccine-induced encephalopathy" began to be a subject of concern. Byers and Moll (1948) described the follow-up status of 15 children

who had developed seizures following vaccination and reported only 1 child neurologically normal 2 years later (Byers & Moll, 1948). Others also noted adverse reactions, but probably because vaccination was optional and of limited availability, these did not arouse public interest.

The serious public controversies regarding pertussis vaccine arose when the first modern whole-cell vaccine, which was combined with vaccines against diphtheria and tetanus, was produced in 1942 (the DTP or DPT vaccine). This vaccine was undoubtedly effective, and there was a very rapid uptake. On the basis of the proven effectiveness and in the context of intensive pharmaceutical lobbying, government intervened in many countries of the world to promote compulsory mass vaccination. By the mid-1960s, for instance, laws had been passed in several states in the United States to ensure the compulsory use of this vaccine in children prior to entering school (Allen, 2007), a logic based on contemporary scientific opinion. This centralist and coercive approach clashed with the principles of parental autonomy. In those days, because of the endotoxin in the early whole-cell vaccines, many vaccinated children developed an immediate minor but alarming febrile reaction. In most this subsided, but in some a febrile seizure occurred. Furthermore, it was also considered that a small number of children developed much more severe neurological complications and in particular what became to be known as *vaccine-related encephalopathy*. This condition has not been well defined, but its core consists of severe epilepsy and mental regression with lifelong disability and dependency. In Britain, where any government policy perceived to be coercive has, since the 17th century, been met with widespread and healthy resistance, the reports of vaccine-induced encephalopathy caused serious public alarm (Kulenkampff et al., 1974) and this prompted the setting up of the "National Childhood Encephalopathy Study" (NCES) in 1976.

The NCES was the first and is still the largest prospective case-control study aimed at assessing the risk of vaccine-induced encephalopathy. The study attempted, through active surveillance to ascertain all children who were admitted to hospital with a severe childhood brain disease, to establish whether there was a temporal association with vaccination. The children were compared to two randomly selected controls matched for age, sex, and area of residence. Outcome was assessed at various time points up to 18 months where a neurological examination and psychometric assessment were carried out. The study provided 5.4 million child-years of observation and over 2 million doses of DTP vaccine were administered during this period. A total of 1182 children were ascertained who developed neurological disease and 39 (3.3%) had done so within 7 days of a DTP vaccination. The attributable risk of a serious neurologic disorder within 7 days of DTP vaccination was reported to be one in 110,000 vaccinations

(Alderslade, 1981). In the original report (Miller et al., 1981), 7 of the 39 had died or were seriously permanently brain damaged. Extrapolating these findings, it was estimated that the rate of permanent neurological deficit after immunization was 1 per 310,000 immunizations (95% confidence interval [CI] 1:531,000–1:54,000), a figure revised to 1 per 330,000 in later publications (Miller et al., 1985). The study emphasized that most cases of neurological disease occurring after vaccination were due to other causes. However, cases of neurological disturbance (of any cause) in the first 72 hours after DTP vaccination were 2.4 times more frequent than in nonvaccinated children. Furthermore, as the rate of neurological disturbance was much lower in DT vaccinated children, this excess risk was considered due to the pertussis component of the vaccine. The findings were criticized on methodological grounds—including selection bias in reporting and in recognition of symptoms in vaccinated cases, recall bias with appreciation of time between vaccination and onset of symptoms, the small number of immunized children, the inclusion of complex febrile seizures and the inclusion of cases where other causes of encephalopathy were present (Scheifele, 1988; Griffith, 1989; Marcuse, 1990). Nevertheless, professional and public debate has raged ever since about these findings.

The public became seriously alarmed and vaccination take-up rates fell dramatically, in Britain to 30% (Brown et al., 2007). Sweden and Germany stopped their vaccine programs altogether and only reinstated them after years of repeated pertussis epidemics. Indeed, in the decade after Sweden withdrew its vaccine (in 1979), Isacson and colleagues (1995) extrapolating from a random selection of 372 10-year olds, suggested that 61% of all Swedish children had contracted pertussis, a depressing and predictable consequence as earlier studies had shown (Royal College of General Practitioners, 1981; Wilson, 1981).

In Britain, the NECS was considered by various regulatory and professional bodies who concluded (rather lamely) that no causal link had been established, but there was strong evidence that such a link might exist in some children (Geier & Geier, 2002). A series of litigation claims were made, but in 1988 a landmark judgment was made in the U.K. high court (the Loveday case; Loveday v. Renton, 1988). In this case, the judge considered first all the evidence as to whether vaccination might cause brain damage and then the specific claims of the plaintiff. In the first phase, which lasted 4 months, a massive examination of the evidence was undertaken, including cross-examination of 19 expert witnesses. The NECS was especially examined and the Judge undertook a case review which concluded that three of the seven children were not brain damaged, two had an illness with viral origins (shown by culture or autopsy), one was a case of Reye's syndrome, and one had probably viral encephalitis on the basis of CSF findings. In this amended form, there was no significant increase in encephalopathy in the vaccinated cases, and in-

deed not a single case could be attributed to vaccination. This massive legal exercise has been widely accepted as the definitive statement on the matter and pretty well ended the litigation, at least in the United Kingdom, in relation to pertussis. This illustrates another truth concerning the relationship of science and social policy, and that is that the latter can often not wait for the former, and by the time science has arrived at a definitive answer, many horses have bolted.

Subsequent large-scale studies have failed to show an association. For instance, Gale et al. (1994) conducted a prospective active surveillance program to assess the feasibility of conducting an NCES-like study in the United States, in a population of 218,000 children under the age of 2 years, and identified a total of 424 cases of neurological illness. Each child was matched to two population control children by birth date, gender, and county of birth. The odds ratio (OR) for encephalopathy or complicated seizures was 3.6 (CI, 0.8–15.2); however, the OR for serious acute neurological illness within 7 days of receipt of DTP vaccine was only 1.1, CI, 0.6–2.0). No elevated risk was observed for nonfebrile seizures (OR, 0.5; 95% CI, 0.2–1.5). The study concluded that there was no statistically significant increased risk of onset of serious acute neurological illness in the 7 days after DTP vaccine exposure for young children, although the power was insufficient to exclude a very rare event. Similarly, in a retrospective case-control study by Ray et al. (2006) among 2,197,000 children, ascertained a total of 452 encephalopathy cases. Of these, 49 had received DTP in the previous 60 days, of which 15 (31%) were of known nonvaccine cause. Six cases of encephalopathy were exposed to DTP within 7 days of onset, which corresponds roughly to an all-cause incidence (not an attributable risk) of one in 370,000, a rate that is not statistically different from the background risk. Similarly, Jackson et al. (2002) found a small risk of febrile seizures (1 per 19,496 vaccinations), which is similar to that from any other cause of fever, the same relatively benign consequences as any other febrile seizure and a very low risk of afebrile seizures (1 case in the 76,133 doses of DTaP vaccine given).

In the United States, similar public disquiet existed, and this was fuelled by a series of high profile law suits, a notorious television program “DTP-vaccine roulette,” and an active parents advocacy group—Dissatisfied Parents Together (DPT). By 1985, 219 suits has been filed against the vaccine manufacturers alleging harm to children due to the whole-cell pertussis vaccine, and one case was settled for \$26 million (Allen, 2007). The mire grew muddier as several doctors made fortunes acting as expert witnesses in high stakes court cases. One was alleged to have received over \$2 million in fees and gifts from pharmaceutical companies (Geier & Geier, 2002). In 1985, the U.S. Institute of Medicine, an independent body one of whose roles is to ensure medical credibility, published its first

panel review. The panel estimated that the 18 million administrations of DTP vaccine had resulted in 7.2 million minor reactions, 10,300 cases of seizures, 164 cases of encephalopathy, 60 cases of chronic disability, and 2–4 deaths per year (U.S. Institute of Medicine, 1985). In 1991, the Institute of Medicine panel met again and concluded its report that the risk of an encephalopathic reaction in children in the first 7 days after DTP vaccination was 3.3 case per million and that 77% of these cases were due to the vaccination (Geier & Geier, 2002). In 1994, the Institute of Medicine published its third report, and in this had shifted its ground. It concluded that now it was not possible to decide whether DTP: (a) caused brain damage, (b) unmasked or triggered an encephalopathy in predisposed persons earlier than would have naturally occurred and thus resulted in brain damage as the earlier the encephalopathy, the more severe the brain damage, or (c) simply triggered an encephalopathy in children who were predisposed in any event. However, the balance of evidence was that there was a causal relationship, albeit a rare one, between DTP and brain damage (U.S. Institute of Medicine, 1994). Gigantic studies are required to refute hypotheses of significant associations of very rare events, reinforcing the adage that proving a negative is virtually impossible.

Another strand to the controversy relates to the differences between the original whole-cell versus acellular vaccine. The original whole cell vaccines contain an endotoxin and are associated with a high frequency of minor reactions. Acellular vaccines are associated with a lower rate of adverse reactions. For example, the rate of febrile seizures within 2 days of vaccination among children <2 years of age was 1 per 19,496 vaccinations, which is much lower than the corresponding rate of 1 per 2835 with the whole-cell vaccine (Cody et al., 1981). The acellular vaccines, however, do not appear to be as effective as the whole cell preparations (Bedford, 2000). Interestingly, acellular vaccines were available in the 1960s, but manufacture then ceased largely because of the perceived risks of the market; the average compensation request by 1985 was \$26 million compared to the total U.S. pertussis vaccine market of only \$2 million (Allen, 2007). Despite the greater cost of the acellular vaccines, they have largely replaced the whole cell vaccines, at least in Western countries, because the public faith in these compounds has grown. This issue remains important, however, in developing countries because of the potentially better cost-effectiveness of whole cell vaccines.

Pertussis vaccine is clearly effective in reducing disease, but pertussis, unlike smallpox, is not yet “eradicated.” In 2000, the WHO reported that pertussis is still one of the top causes of vaccine-preventable deaths, with over 300,000 children dying per year worldwide (Crowcroft & Britto, 2002), and the number of reported cases continues to rise. One reason is inadequate herd immunity, and the disease is increasing particularly in older children and young adults,

in whom the effects of vaccination wear off, and who thus act as reservoirs of disease for the vulnerable population of neonates and very young infants who are at risk before vaccination. This has led to the recommendation that adults and teenagers should have booster vaccination (Crowcroft et al., 2003; Schafer et al., 2006). Current U.K. guidelines are that acellular pertussis vaccine should be given (in an injection combined with diphtheria, tetanus, *Haemophilus influenzae* type b, and inactivated polio vaccines) at 2, 3, and 4 months of age and then a booster (given with inactivated polio vaccine) at between 3 years 4 months and 5 years of age. Existing epilepsy and/or a stable neurological condition are not considered contraindications.

The debate about the potentially harmful effects of vaccination has now been moved on by the publication of a very important study by Berkovic et al. (2006). As part of a study of unexplained encephalopathies with seizures occurring during the first year of life, the authors identified cases from child neurologists in Australia, New Zealand, Canada, and Scotland. Of 96 cases, 14 had the onset of symptoms within 72 hours of pertussis vaccination. Although no restrictions were placed on acceptance into the study other than that the encephalopathy be unexplained by other causes (e.g., infection, malformation, metabolic disorder, etc.), 11 of those who met criteria for postvaccine encephalopathy corresponded to the phenotype for severe myoclonic epilepsy of infancy (SMEI or Dravet syndrome Dravet et al., 2005). SMEI is often caused by severe and often spontaneous mutations in the *SCN1A* gene (Claes et al., 2001; Harkin et al., 2007). The investigators found that 11 of the 14 children with presumed vaccine-related encephalopathy, and 10 of the 11 with the SMEI phenotype, had mutations in this gene. These mutations were not present in the normal population and were demonstrated to be spontaneous in the 9 cases for whom parental DNA was available. The presence of these mutations provides a compelling explanation of the cause of the encephalopathy. In the other three cases, no genetic or other cause was found. We caution, however, that in many monogenic disorders, mutations are not the only mechanisms for genetic error. Increasingly, deletions, not detected by standard mutational analyses, are being found in the same genes (e.g., SMEI (Mulley et al., 2006), Rett syndrome (Percy, 2007)). Copy number analysis in the remaining three cases did not reveal deletions in the *SCN1A* gene (Scheffer, personal communication); however, testing for deletions should still be considered in future studies of this issue. These findings raise the possibility that many cases of apparent vaccine-encephalopathy could in fact be cases of SMEI, and due to an inherent genetic defect with no causal relationship with vaccination. This is the first study to demonstrate this, and replication would be a welcome addition to the scientific literature. Even though the results are currently unreplicated, they are so striking that this must be considered a hammer blow to the repeatedly unsubstantiated

hypothesis that pertussis vaccination is the cause of epileptic encephalopathy (Berg, 2007; Brown et al., 2007).

Another interesting question is raised, in our opinion, which might prove a fruitful area for future research. Could encephalopathy following early pertussis infection itself be due not to the infection but to the same underlying *SCN1A* genetic defect (unmasked by the accompanying fever)? If this were the case, then the currently accepted estimate of the risk of pertussis-induced encephalopathy may also be too high. This has not been investigated but we feel it is important to do.

The Australian study has moved the debate on, and it is worth summing up where we are now in this prolonged saga which has spanned at least 70 years, and which has generated vast heat and smoke, fortunes for lawyers and witnesses, and prolonged misery for families.

1. Whooping cough can cause encephalopathy and death in neonates and young children. The attributable risk of encephalopathy due to pertussis is currently estimated to be between 1 in 1,200–12,000 infections—much higher than any estimate of the risk of DTP vaccination; the risk is highest in neonates and infants. However, it is conceivable, although not studied, that some of these cases might have the same genetic basis as the alleged vaccine cases.
2. Pertussis vaccination is a highly effective means to prevent a disabling and sometimes deadly disease. It provides excellent protection to the young children vaccinated and also “herd immunity” which is vital in the protection of neonates, who might be at risk before their vaccination is due, and of immunocompromised individuals. Adding booster doses to older children is increasingly advocated for this purpose, and in countries where vaccination rates are high and where booster doses are administered, rates of pertussis are very low.
3. Side effects of vaccination include fever (particularly common following the whole cell vaccine). As vaccination is given at an age when seizures due to fever occur, some febrile seizures occur, mediated via fever rather than by a direct cerebral action of the vaccine itself. It seems that vaccine-induced febrile seizures are no more frequent than febrile seizures due to any other cause of fever and carry the same relatively benign consequences. The rate of fever and also thus of febrile seizures after the acellular vaccines is less, especially in older children.
4. The rate of nonfebrile seizures associated with vaccination is very low (an example estimate is 1 per 76,133 vaccinations; Jackson et al., 2002).
5. The extremely emotive debate that has raged about whether pertussis vaccination can cause encephalopathy continues to smolder. The NCES and

other earlier studies purporting to show an association have been the subject of detailed criticisms that invalidate their conclusions in our view, and more recent studies have failed to demonstrate any causal association between vaccination and the occurrence of encephalopathy. It is always difficult to prove a negative, but the evidence is convincing enough now for most of the scientific and medical community (including the authors) to conclude that the vaccines are generally safe. Even among those who believe that encephalopathy does occur, all accept that this must be a rare event. Even the worst estimates suggest a frequency of less than 3 per million vaccinations; figures which are much lower than the risk of encephalopathy due to pertussis.

6. Vaccine-induced encephalopathy (or epilepsy) can be diagnosed only when other forms of childhood encephalopathy are excluded. The recent Australian finding reported by Berkovic et al. (2006) shows that a proportion of cases previously attributed to vaccine encephalopathy are in fact due to the genetically determined SMEI. It is possible that the vaccine causes a fever, which precipitates the manifestations of this condition—in other words brings its occurrence forward in time—but the vaccine cannot be considered to be the primary cause in these cases.
7. Any child in the future suspected as having a vaccine-induced encephalopathy (or epilepsy) must be tested for SMEI before the diagnosis can be accepted, at least in scientific and presumably for the purposes of any litigation proceedings.

Despite the well-documented benefits and safety of vaccination, the cases of the occasional child who develops an encephalopathic condition or epilepsy in close temporal association with receipt of vaccine continues to create despair and anger. In reaching the conclusion that vaccines do not cause encephalopathy (except possibly extremely rarely), we do not mean to ignore or minimize the tragedy for families with children who have developed an epileptic encephalopathy. The recent Australian findings are important in this regard. Knowing that the assumed culprit, the pertussis vaccine, was not the cause but rather a spontaneous genetic mutation is responsible, and thus that the parents could not have prevented the illness by withholding vaccine, should lessen just a little the feelings of responsibility and guilt. It must also be recognized that the finding of a SMEI mutation may well mean that financial compensation under any vaccine compensation scheme that depends on the demonstration of a causal relationship will not be available; although, in the absence of a *SCN1A* errors and unless other genetic mechanisms are established, legal claims regarding vaccine encephalopathy in specific cases may not be deterred or may even be encouraged.

If properly communicated, the Australian findings should also do much to improve the public's understanding of the true risks and the safety of this vaccine. Such an understanding cannot come too soon as, in both the United Kingdom and United States, a new legal cause is being prosecuted, this one aimed at inculpating MMR vaccine (in the United Kingdom) and particularly a preservative previously used in vaccines, thimerisol (in the United States), as the cause(s) of autism. The parallels are striking. The MMR vaccine prevents measles, another infectious disease with devastating consequences including progressive encephalopathy and death. Scientific studies at this stage provide conflicting results, but the weight of current evidence indicates no increased risk of autism. This is very similar to the situation in relation to pertussis vaccination in the 1980s and 1990s. Scientific discovery rarely keeps pace with the need for knowledge in setting rational public policy, but we can hope, however, that we as a society learn from the pertussis vaccine-induced injury-sodium channel story and do not rush to premature judgment.

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